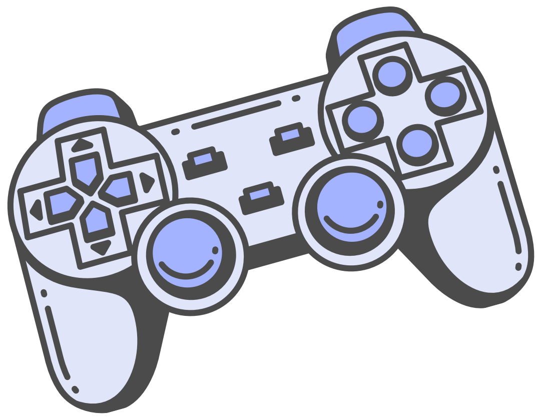
File Handling in Python

Introduction to reading, writing, and managing files in Python

By:-
Harsh Modi 202501100136
SE2(S22)

File Handling in Python

Managing data stored in files using Python. Enables storing data, logs, and reports persistently. Real-world uses include saving program output, reading configuration files, and data analysis.



File Modes in Python

r
Read mode (open file for reading)

Mode

w
Write mode (open file for writing, overwrite)

a
Append mode (add to end of file)

x
Create mode (create new file, fails if exists)

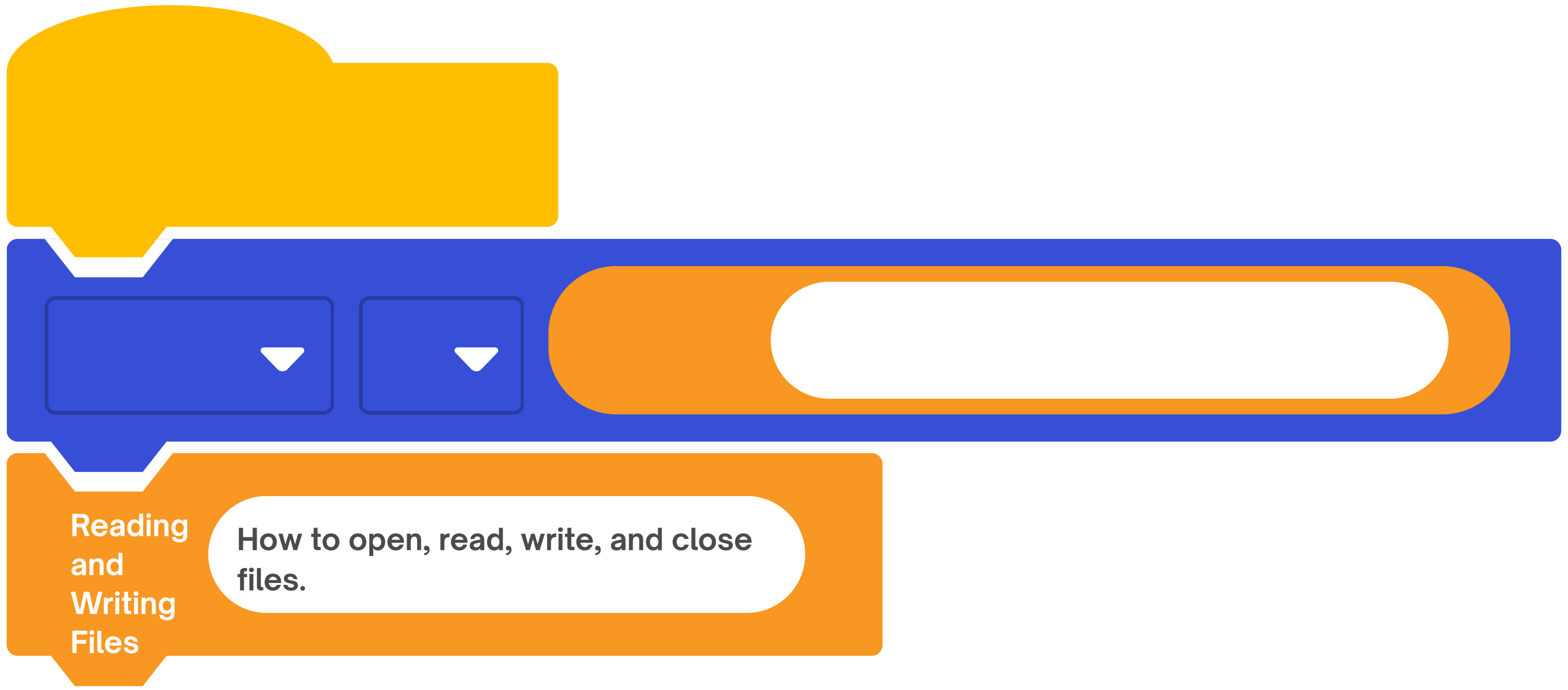
rb / wb
Binary read/write modes (for non-text files)

Tells a program what kind of value a variable can hold.

text in quotes

(e.g. “hello”)

```
file = open('example.txt', 'r')
content = file.read()
file = open('example.txt', 'w');
file.write('Hello, world!')
file.close()
```



Using with Statement & Exception Handling

The with statement automatically closes files after use – safer and cleaner. try-except handles errors like FileNotFoundError.

```
try:
with open('example.txt', 'r') as file:
content = file.read()
except FileNotFoundError:
print("File not found.")
```

Advantages of File Handling in Python

- Easy to store and retrieve data persistently
- Supports various file modes for flexibility
- with statement ensures safe file management
- Exception handling improves program robustness

Thank you for your attention!

Practice file handling with Python!